

Hole-defect pairing and independent realization for spectra of restrictions

Candidate partial answer to the final question in arXiv:2004.11708

11 August 2026

Status: candidate partial result. For an arbitrary bounded operator and invariant subspace, a canonical exact defect pairing identifies every filled resolvent component with an open set of eigenvalues of the quotient operator; the common defect is constant throughout the component. Conversely, one fixed nonnormal operator with finitely many circular holes realizes every fill/no-fill pattern, and even every prescribed finite-or-infinite defect-multiplicity profile, by varying its invariant subspace.

1 Source question and result

Let $T \in \mathcal{B}(X)$ on a complex Banach space and let $M \subset X$ be a closed T -invariant subspace. Write

$$T_M = T|_M, \quad \dot{T}(x + M) = Tx + M \quad (x \in X).$$

Drivaliaris and Yannakakis [1] recall that a bounded component D of $\rho(T)$ is either disjoint from $\sigma(T_M)$ or is entirely contained in it. Their final remark asks which holes can be filled. The following theorem gives a defect-theoretic answer for a fixed pair (T, M) and records the exact complementary phenomenon on the quotient.

For an operator with closed range, call

$$\text{def } A = \dim(Y/\text{ran } A)$$

its defect, with the value ∞ when the quotient is infinite-dimensional.

Theorem 1 (Canonical hole-defect pairing). *For every $\lambda \in \rho(T)$:*

1. $T_M - \lambda$ is bounded below and $\dot{T} - \lambda$ is surjective;
2. there is a canonical Banach-space isomorphism

$$\delta_\lambda : \ker(\dot{T} - \lambda) \longrightarrow M/(T_M - \lambda)M, \quad \delta_\lambda(x + M) = (T - \lambda)x + (T_M - \lambda)M; \quad (1)$$

3. on each connected component D of $\rho(T)$, the common extended dimension

$$d_D = \dim \ker(\dot{T} - \lambda) = \text{def}(T_M - \lambda) \in \mathbb{N}_0 \cup \{\infty\}$$

is independent of $\lambda \in D$.

Consequently,

$$D \subseteq \sigma(T_M) \iff d_D > 0 \iff D \subseteq \sigma_p(\dot{T}).$$

When these conditions hold, D lies in the residual spectrum of T_M .

Thus a filled hole is not merely extra spectrum of a restriction: point by point it is an eigenvalue region of the quotient, and its quotient eigenspace is canonically paired with the cokernel of the restriction. The residual spectrum conclusion itself is classical; see, for example, Barnes [2, Results 1]. The explicit isomorphism (1), its componentwise defect consequence, and the realization below are the useful combined formulation here.

2 Proof intuition

Outside $\sigma(T)$ the operator $T - \lambda$ is invertible. Restricting an invertible map to an invariant subspace cannot create a kernel, but it can lose surjectivity. Exactly the same loss reappears on the quotient in the opposite form: the induced quotient map is always onto, but can acquire a kernel. Applying $(T - \lambda)^{-1}$ turns a missing range class in M into the corresponding quotient eigenvector, which gives (1).

The restriction remains bounded below everywhere in $\rho(T)$, so it stays in the upper semi-Fredholm set. Its index is locally constant there and hence constant on every resolvent component. This upgrades the usual all-or-none spectral statement to a constant multiplicity statement.

For realization, a bilateral shift supplies a circular spectrum and its unilateral invariant half supplies the filled disk. Countably many copies allow an arbitrary defect multiplicity in each disk. A 2×2 Jordan extension makes the ambient operator nonnormal without changing either its spectrum or the restrictions used in the construction.

3 Proof of Theorem 1

Fix $\lambda \in \rho(T)$. For $m \in M$,

$$\|(T_M - \lambda)m\| = \|(T - \lambda)m\| \geq \frac{1}{\|(T - \lambda)^{-1}\|} \|m\|.$$

Thus $T_M - \lambda$ is injective with closed range. If $y + M \in X/M$ and $x = (T - \lambda)^{-1}y$, then

$$(\dot{T} - \lambda)(x + M) = y + M,$$

so $\dot{T} - \lambda$ is surjective.

If $x + M \in \ker(\dot{T} - \lambda)$, then $(T - \lambda)x \in M$, so (1) makes sense. Replacing x by $x + m$ changes the displayed representative by $(T_M - \lambda)m$, hence δ_λ is well defined. Its inverse is the equally explicit map

$$y + (T_M - \lambda)M \mapsto (T - \lambda)^{-1}y + M. \quad (2)$$

Both maps are bounded, because they are induced by bounded operators between quotient spaces. Equations (1) and (2) are inverse to one another, proving the canonical isomorphism.

Every $T_M - \lambda$, $\lambda \in \rho(T)$, is upper semi-Fredholm: it has closed range and zero-dimensional kernel. The semi-Fredholm index is locally constant on the upper semi-Fredholm set, with the extended value $-\infty$ when the defect is infinite. Since

$$\text{ind}(T_M - \lambda) = -\text{def}(T_M - \lambda),$$

connectedness gives the asserted constant d_D . Finally, $\lambda \in \sigma(T_M)$ is equivalent to failure of surjectivity, because the restriction is already bounded below. This is equivalent to $d_D > 0$, and by (1) to $\ker(\dot{T} - \lambda) \neq 0$. A proper closed range is not dense, proving the residual-spectrum assertion. $\square$

4 One nonnormal operator realizing all finite hole profiles

The preceding invariant has no universal interaction constraint across different holes, even if the ambient operator is fixed and nonnormal.

Theorem 2 (Independent realization with prescribed defects). *Let*

$$\overline{D}_j = \{z \in \mathbb{C} : |z - c_j| \leq r_j\}, \quad 1 \leq j \leq m,$$

be pairwise disjoint closed disks, and put $\Gamma_j = \partial D_j$. There is a single nonnormal operator J on a separable Hilbert space with

$$\sigma(J) = \bigcup_{j=1}^m \Gamma_j$$

such that, for every $(d_1, \dots, d_m) \in (\mathbb{N}_0 \cup \{\infty\})^m$, there is a closed J -invariant subspace $M_{\mathbf{d}}$ satisfying

$$\sigma(J|_{M_{\mathbf{d}}}) = \bigcup_{d_j=0} \Gamma_j \cup \bigcup_{d_j>0} \overline{D}_j. \quad (3)$$

Moreover, the constant defect from Theorem 1 on D_j is exactly d_j .

Proof. Let U be the bilateral shift on $\ell^2(\mathbb{Z})$ and let $H_+ = \overline{\text{span}}\{e_n : n \geq 0\}$. Thus H_+ is U -invariant, $U|_{H_+}$ is the unilateral shift S , and

$$\sigma(c_j I + r_j U) = \Gamma_j, \quad \sigma(c_j I + r_j S) = \overline{D}_j.$$

On

$$G = \bigoplus_{j=1}^m \bigoplus_{k=0}^{\infty} \ell^2(\mathbb{Z})$$

define the normal operator B to act as $c_j I + r_j U$ on the (j, k) copy. Then $\sigma(B) = \bigcup_j \Gamma_j$. On $G \oplus G$, set

$$J = \begin{pmatrix} B & I \\ 0 & B \end{pmatrix}.$$

The nilpotent off-diagonal part commutes with the diagonal, so $\sigma(J) = \sigma(B)$. Also J is not normal: the top-left blocks of J^*J and JJ^* differ by I .

For the profile $\mathbf{d}$, define a closed B -invariant subspace $N_{\mathbf{d}} \subset G$ as follows. In every j -family include the entire $k = 0$ copy. In precisely d_j of the remaining copies include H_+ (all remaining copies when $d_j = \infty$), and put zero in the other copies. Finally set

$$M_{\mathbf{d}} = N_{\mathbf{d}} \oplus \{0\} \subset G \oplus G.$$

It is J -invariant and $J|_{M_{\mathbf{d}}}$ is exactly $B|_{N_{\mathbf{d}}}$. The full $k = 0$ copy contributes Γ_j ; each included half-space contributes $\overline{D}_j$. This proves (3).

If $\lambda \in D_j$, then $c_j I + r_j U - \lambda$ is invertible, while $c_j I + r_j S - \lambda$ is bounded below with range of codimension one. All blocks belonging to the other disks are invertible because the closed disks are disjoint. Direct sums therefore give defect d_j , as claimed. $\square$

5 Scope and novelty caution

Theorem 1 gives an exact structural test once the invariant subspace is fixed, and Theorem 2 shows that arbitrary finite profiles occur without cross-hole restrictions for one nonnormal operator. This does *not* classify which profiles occur in the invariant-subspace lattice of an arbitrary prescribed operator. That remaining question can encode difficult information about the lattice itself.

The individual ingredients are classical in flavor. In particular, the all-or-none theorem and the residual-spectrum statement predate the source, Barnes [2] studies restriction/quotient Fredholm relations and gives an invertible shift/backward-shift triangular example, and Conway–Olin solve the normal-Hilbert case [3]. Bounded exact-phrases and subject searches through 11 August 2026 did not locate the precise canonical pairing plus one-fixed-nonnormal-operator simultaneous multiplicity realization stated above. The appropriate novelty claim is therefore a candidate partial synthesis, not a claimed solution of the full arbitrary-operator problem.

References

- [1] D. Drivaliaris and N. Yannakakis, *The spectrum of the restriction to an invariant subspace*, Operators and Matrices 14 (2020), 261–264; arXiv:2004.11708.
- [2] B. A. Barnes, *Spectral and Fredholm theory involving the diagonal of a bounded linear operator*, Acta Sci. Math. (Szeged) 73 (2007), 237–250.
- [3] J. B. Conway and J. E. Olin, *A functional calculus for subnormal operators II*, Mem. Amer. Math. Soc. 10, no. 184 (1977).